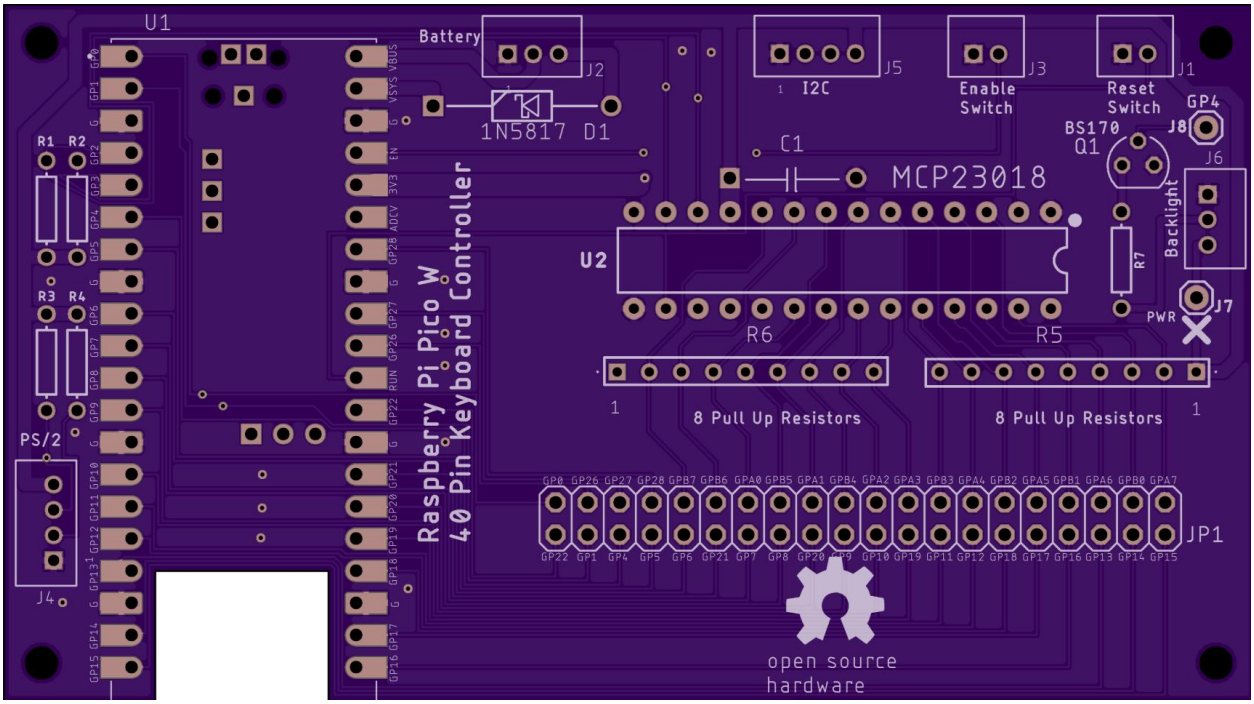
Raspberry Pi Pico W with Port Expander

I have designed a laptop keyboard controller board using a Raspberry Pi Pico (W) that includes an MCP23018 port expander. This increases the 26 pin GPIO limit of a stand-alone Pi Pico to 40 GPIO's (16 from the port expander and 24 from the Pico). All components for this board including the FPC connector are through-hole for easy soldering. The Eagle board and schematic files along with a zipped Gerber layout file and software are at my repo.

The OSHPark rendition of the Raspberry Pi Pico W port expander board is shown below:

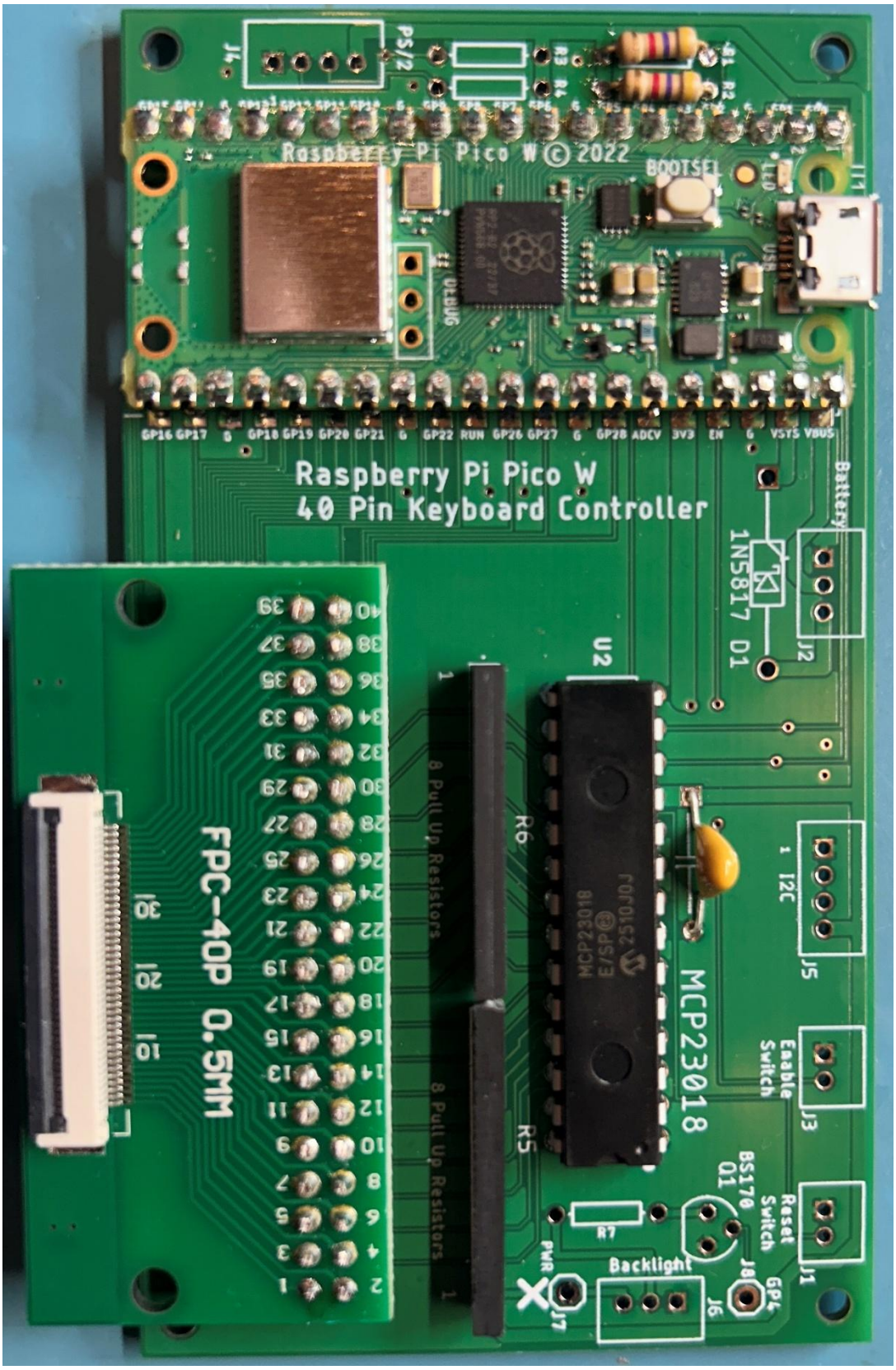
The size is 99mm x 55mm. JLCPCB.com cost is $6 for 5 boards (with slow shipping).



The cutout in the board gives better reception for the Bluetooth antenna on the Pico. The Pico can be mounted with header pins or soldered directly to the board for a lower profile.

The two rows of 20 pins at JP1 are for mounting an FPC adapter board. Choose a board with the correct pitch and pin count that matches your keyboard's FPC cable. Don't get a board with pre-soldered header pins or you'll have to mount it underneath the Pico board. Solder your own header pins with the adapter board sitting on top. Odds are your cable won't have 40 pins so install the FPC adapter board as far to the right as possible, leaving the pads on the left side empty. This will free up Pico GPIO pins for other uses such as a touchpad and backlight control.

This picture shows the assembled board (without the extra items).



Bluetooth Low Energy (BLE) operation implies the controller will be running from a separate lithium battery with a charging circuit. USB 5 volts on the Pico VBUS pin is routed to pin 2 of a 3 pin JST connector at J2. This should be cabled (along with ground) to an off-board battery charger circuit. The nominal 3.7 volt battery output should be cabled to J2 pin 3. This feeds the anode of a 1N5817 Schottky diode at D1 that "or-ties" the battery voltage to the Pico's VSYS pin. When the USB cable is attached, a Schottky diode inside the Pico brings USB power to VSYS. If the keyboard will only be used for USB, it will always have power so Schottky diode D1 and the J2 connector are not needed.

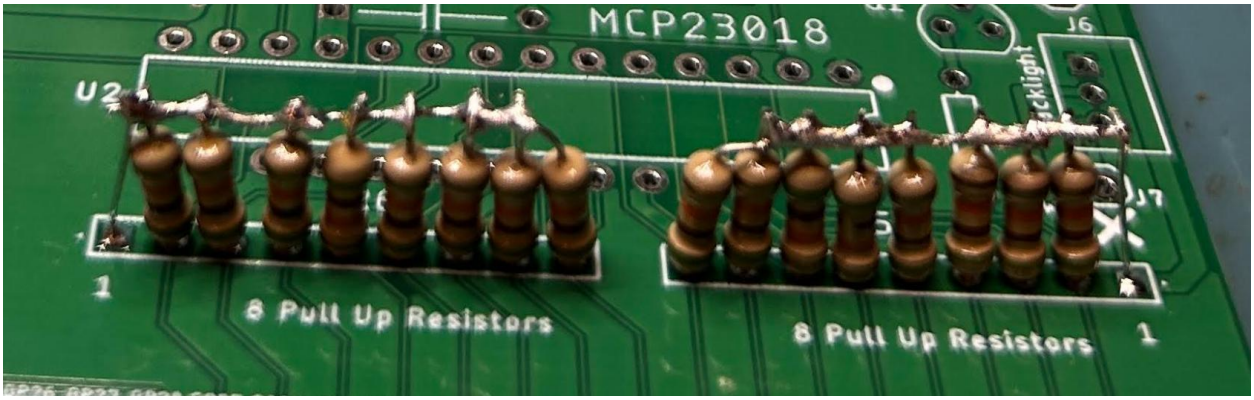
The EN signal at Pico Pin 37 is routed to a 2 pin JST connector at J3. If you want to power down the Pico to save the battery, add a switch to connect the EN signal to ground.

Resistors R1 and R2 are the I2C clock and data pullups and should be 4.7KΩ for running the bus at 400KHz. The I2C bus controls the MCP23018 port expander and it is also routed to the 4 pin JST connector at J5. This connector is only needed if interfacing to an I2C touchpad.

Clock and data pullup resistors R3 and R4 are 4.7KΩ and are driven by GPIO 0 and 1 of the Pico. These resistors and the 4 pin JST connector at J4 are only needed if interfacing to a PS/2 touchpad. Note that using these GPIO's for PS/2 reduces the available keyboard FPC pins to 38 instead of 40.

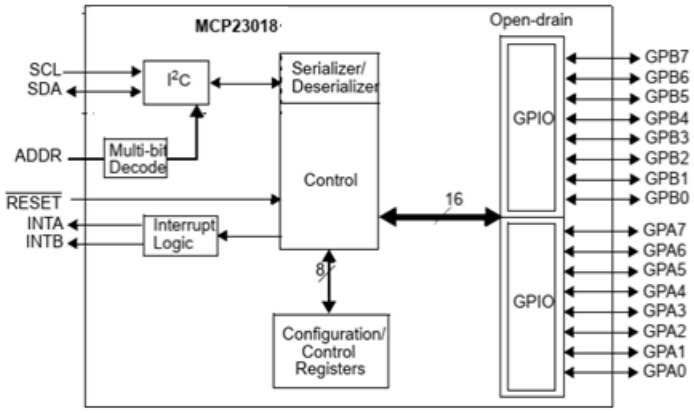
An additional keyboard pin can be repurposed for controlling a backlight LED by installing a 3 pin JST connector at J6. This connector provides 3.3 volts, ground, and an open drain N Channel FET with a current limit resistor for controlling an LED. Many laptop keyboards provide the backlight LED anode and cathode on multiple pins of a small FPC cable. Use a small FPC adapter board to wire the anode pins to 3.3 volts at J6 pin 3 and the cathode pins to J6 pin 2. Install a current limit resistor at R7 and a BS170 NFET at Q1. Size the resistor so the current at max brightness stays well under the Pico's 300ma limit. Pico GPIO 4 can be programmed to output a PWM signal to the gate of the NFET which turns on and off the high current to the backlight LED.

The MCP23018 Port Expander has 16 I/O pins. When programmed to be an output, it has open drain FETs instead of the typical totem pole configuration found in its sister part, the MCP23017. The MCP23018 can only drive low or float the output for a high, relying on a pull up resistor elsewhere in the circuit. Laptop keyboards do not have diodes on each switch so it's important to not create a sneak path to power when multiple keys are pressed. When programmed to be an input, there are internal pull up resistors that can be turned on in the code. Unfortunately, those resistors are about 100KΩ which is too high a value to quickly pull up the column pins of the key matrix. I added pads on the circuit board for 16 pull up resistors. You can either use individual resistors as shown below or use two 9 pin pull up packages like the 4309R-101-473LF.



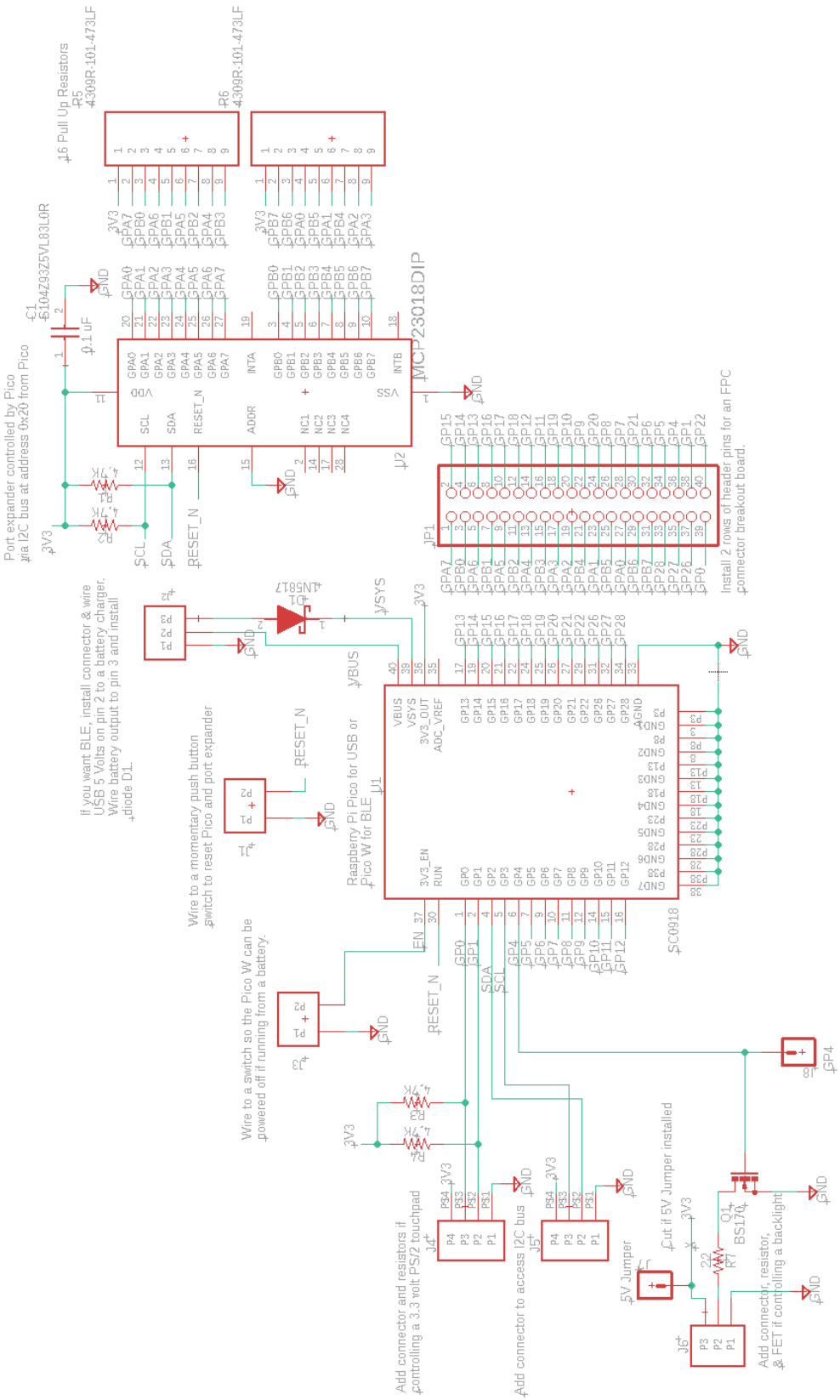
The resistor value determines how quickly the signal returns high when a key is released. You can add some delay in the code before reading the key matrix but too big a delay will be noticeable. The Pico has programmable pull up resistors that are about 50KΩ. I've done a lot of testing with Pico controlled keyboards and the 50KΩ pullup value has not caused any problems.

The MCP23018 port expander diagram is shown below.



My board powers this chip from the 3.3 volt regulator at Pico pin 36. The SDA and SCL signals are driven from the Pico I2C pins at GP2 and GP3. The 4.7KΩ pullup resistors (and short traces) allow the I2C bus to operate at 400KHz. The Address pin is grounded to set the I2C address for the port expander at 0x20. The MCP23018 RESET_n pin is tied to the Pico's RUN pin and to a 2 pin JST connector at J1 that can be wired to a push button reset switch. The INTA and INTB pins are not used. Each GP pin of the port expander can be programmed over I2C as an input or an output.

The Eagle Schematic is given below:



Programming – This board can be programmed for BLE operation using Arduino C++ code and the Earle Philhower library. The Adafruit TinyUSB library is used to program C++ code for USB operation. QMK can also be used because it has modules for controlling the MCP23018 Port Expander but so far, I have not been able to get the code to compile, (I'll keep trying). Unfortunately, this board will not work with KMK because it does not support port expanders.

The MacBook Pro A1286 keyboard shown below was used for testing the port expander board and developing the BLE and USB software. It has a 30 pin FPC cable with a 0.5mm pitch.

This keyboard only has 30 FPC pins, and I've slid the FPC cable to the far right of the 40 pin connector so that all 16 GPIO's from the port expander are used and the remainder are connected to the Pico GPIO's.

I developed Circuit Python code called "matrix_decoder_mcp23018_40pin.py". This code is similar to the matrix decoder code I've written for the 26 pin board except it controls the port expander over the I2C bus. A feature of this code is that it initially scans all 40 pins and if any are shorted, they are removed from the scan list. The result of running this code on the MacBook keyboard can be found at my repo. I've previously used this keyboard with a Teensy so I was able to check the results for accuracy.

I converted this connection list into the row/column key matrix shown below. Out of the 24 Pico GPIO's, this keyboard used 12. They are denoted by GP followed by a number. The port expander pins are denoted with GPA or GPB corresponding to the A and B ports. Each port has 8 bits numbered 0 through 7.

MacBook Pro A1286 Key Matrix

GPIO #	GPA0	GP7	GPB5	GP8	GP17	GPB1	GP16	GPA6	GPB0
GPA1	q	1	a	z	F11	bckspc	F12	F1	
GP20	w	2	s	x	esc	tab	`	F2	
GPB4	e	3	d	c				F3	
GP9	y	6	h	n	/	space	=	F6	
GPA2	r	4	f	v			]	F4	
GP10					enter			Shift-R	
GPA3				GUI-L	0		GUI-R		
GP19	t	5	g	b	-	Fn	[	F5	
GPB3					F10	Alt-R			
GP11					p	Caps lk			
GPB2	u	7	j	m	;	quote		F7	
GP18	i	8	k	comma		left	right	F8	
GPA5	o	9	L	period	\	down	up	F9	
GP14									Shift-L
GPA7									Alt-L
GP15									Cntr-L

The "Pico_W_MacBookA1286_BLE .ino" code cycles through all possible GPIO's in the Pico and port expander using an outer row loop with an inner column loop (a loop inside a loop). The Pico GPIO bits are denoted as 0, 1, 4 – 22, 26 – 28. There are missing GPIO's because 2 and 3 are used for I2C and 23 – 25 are for internal Pico use only. In the port expander, the first 8 bits are GPA pins and denoted as 100 – 107 in the code. The second 8 bits are GPB pins and denoted as 108 - 115. The code can tell when it needs to control a port expander bit as a row or column by checking if it's >= 30. This table maps the 100 – 115 bit numbering to the GPA and GPB registers in the port expander.

GPA0	GPA1	GPA2	GPA3	GPA4	GPA5	GPA6	GPA7	GPB0	GPB1	GPB2	GPB3	GPB4	GPB5	GPB6	GPB7
100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115

Even though this keyboard has a 30 pin FPC connector, only 25 pins are for the keyboard keys. The remainder are no-connects, grounds, or LED's.

To change the BLE code to work with a different keyboard, you would first run the matrix decoder Python code to make a connection list. Then either build a key matrix by hand or ask your favorite AI assistant (like ChatGPT, Claude, or Gemini) to analyze your filled-out keyboard pin list text file. To force the AI to accurately find the hidden matrix geometry without creating a giant, un-optimized grid, copy and paste the following prompt.

AI Prompt: "I am building a laptop keyboard controller using a Raspberry Pi Pico and the Earle Philhower C++ Arduino BLE libraries. I have attached my raw keyboard GPIO pin connection list

below. Because laptop matrices lack schematics and omit diodes, I do not know the exact number of rows and columns. Analyze the overlapping intersections in my pin list to isolate the optimized, condensed hardware matrix geometry. The column axis will represent a small group of shared lines (typically 8 or 9 pins), while the intersecting row axis will hold the remaining unique pins. Organize these into an efficient row_pins and col_pins configuration (do not output a giant, un-optimized square array). Here is my raw pin connection file: ***[INSERT YOUR FILE HERE or PROVIDE A LINK]****".*

It is also possible to ask the AI to modify my BLE code with your keyboard's matrix information but if you want to do the modifications yourself, follow these steps:

Update the number of row and column pins at lines 25-26:

#define NUM_ROWS 16

#define NUM_COLS 9

List the row and column pins at lines 33 & 37 using the port expander translation table above:

```
const uint8_t row_pins[NUM_ROWS] = {
  101, 20, 112, 9, 102, 10, 103, 19, 111, 11, 110, 18, 105, 14, 107, 15
};
const uint8_t col_pins[NUM_COLS] = {
  100, 7, 113, 8, 17, 109, 16, 106, 108
};
```

Update the key matrix table shown below (lines 40 – 58). Note the letters, numbers, and punctuation are inside single quotes but other keys use KEY_. Backspace and quote are special:

'\\' this is the backspace key

'\'' this is the single quote key

```
const uint8_t keymap[NUM_ROWS][NUM_COLS] = {
//  GPA0=100    7           GPB5=113    8           17              GPB1=109        16              GPA6=106        GPB0=108
  { 'q',        '1',        'a',        'z',        KEY_F11,        KEY_BACKSPACE,  KEY_F12,        KEY_F1,         0 },            //GPA1=101
  { 'w',        '2',        's',        'x',        KEY_ESC,        KEY_TAB,        '`',            KEY_F2,         0 },            //20
  { 'e',        '3',        'd',        'c',        0,              0,              0,              KEY_F3,         0 },            //GPB4=112
  { 'y',        '6',        'h',        'n',        '/',            ' ',            '=',            KEY_F6,         0 },            //9
  { 'r',        '4',        'f',        'v',        0,              0,              ']',            KEY_F4,         0 },            //GPA2=102
  { 0,          0,          0,          0,          KEY_RETURN,     0,              0,              KEY_RIGHT_SHIFT,0 },            //10
  { 0,          0,          0,          KEY_LEFT_GUI,'0',           0,              KEY_RIGHT_GUI,  0,              0 },            //GPA3=103
  { 't',        '5',        'g',        'b',        '-',            0,              '[',            KEY_F5,         0 },            //19
  { 0,          0,          0,          0,          KEY_F10,        KEY_RIGHT_ALT,  0,              0,              0 },            //GPB3=111
  { 0,          0,          0,          0,          'p',            KEY_CAPS_LOCK,  0,              0,              0 },            //11
  { 'u',        '7',        'j',        'm',        ';',            '\'',           0,              KEY_F7,         0 },            //GPB2=110
  { 'i',        '8',        'k',        ',',        0,              KEY_LEFT_ARROW, KEY_RIGHT_ARROW,KEY_F8,         0 },            //18
  { 'o',        '9',        'l',        '.',        '\\',           KEY_DOWN_ARROW, KEY_UP_ARROW,   KEY_F9,         0 },            //GPA5=105
  { 0,          0,          0,          0,          0,              0,              0,              0,              KEY_LEFT_SHIFT }, //14
  { 0,          0,          0,          0,          0,              0,              0,              0,              KEY_LEFT_ALT },   //GPA7=107
  { 0,          0,          0,          0,          0,              0,              0,              0,              KEY_LEFT_CTRL }   //15
};
```

Change the Bluetooth name associated with your keyboard at line 214:

KeyboardBLE.begin("MacBook Pro HID", "Apple Wireless Mod");

In the Arduino IDE, make sure to set the IP/Bluetooth Stack to IPV4 + Bluetooth. Any compiler errors can be fed to AI along with a link to the code and it will tell you what to fix. This also applies if the code compiles but fails to work correctly. If you're having trouble getting Bluetooth to connect to your PC or phone, but it worked previously. Remember that each time you re-flash the Pico, you need to "Remove Device" in your PC or phone's Bluetooth settings and then pair it again.

If you want the media keys to work, look at lines 248 through 284. Modify the row and column index numbers so it can detect when the Fn key is held down and when the desired Function key is pressed.

Arduino C++ USB Code

You will need to load the Adafruit TinyUSB Library into the Arduino IDE. Under Tools, USB Stack, select Adafruit TinyUSB. The example USB code for a Macbook Pro A1286 laptop keyboard is at my repo. Modify this code for your keyboard by determining its key connections using the Circuit Python code at my repo and then building the key matrix (either manually or with the help of AI). Modify the C++ code with your key matrix and row/column GPIO numbers. The sections you will need to change are shown below.

// Row and Column count

const int NUM_ROWS = 16; // number of row signals

const int NUM_COLS = 9; // number of column signals

The I/O pins are numbered differently depending on if they are connected to the Pico or the Port Expander. GPIO bits denoted as 0, 1, 4 – 22, 26 – 28 are going to the Pico. The port expander's first 8 bits are GPA pins and denoted as 100 – 107 in the code. The second 8 bits are GPB pins and denoted as 108 - 115.

// Row pins stored in an array

const int rowPins[NUM_ROWS] = {101, 20, 112, 9, 102, 10, 103, 19, 111, 11, 110, 18, 105, 14, 107, 15};

// Column pins stored in an array

const int colPins[NUM_COLS] = {100, 7, 113, 8, 17, 109, 16, 106, 108};

```
const uint8_t keyMap[NUM_ROWS][NUM_COLS] = {
//  GPA0=100    7           GPB5=113    8               17              GPB1=109            16                  GPA6=106            GPB0=108
  { HID_KEY_Q,  HID_KEY_1,  HID_KEY_A,  HID_KEY_Z,      HID_KEY_F11,    HID_KEY_BACKSPACE,  HID_KEY_F12,        HID_KEY_F1,         0 },                //GPA1=101
  { HID_KEY_W,  HID_KEY_2,  HID_KEY_S,  HID_KEY_X,      HID_KEY_ESCAPE, HID_KEY_TAB,        HID_KEY_GRAVE,      HID_KEY_F2,         0 },                //20
  { HID_KEY_E,  HID_KEY_3,  HID_KEY_D,  HID_KEY_C,      0,              0,                  0,                  HID_KEY_F3,         0 },                //GPB4=112
  { HID_KEY_Y,  HID_KEY_6,  HID_KEY_H,  HID_KEY_N,      HID_KEY_SLASH,  HID_KEY_SPACE,      HID_KEY_EQUAL,      HID_KEY_F6,         0 },                //9
  { HID_KEY_R,  HID_KEY_4,  HID_KEY_F,  HID_KEY_V,      0,              0,                  HID_KEY_BRACKET_RIGHT, HID_KEY_F4,      0 },                //GPA2=102
  { 0,          0,          0,          0,              HID_KEY_ENTER,  0,                  0,                  HID_KEY_SHIFT_RIGHT, 0 },               //10
  { 0,          0,          0,          HID_KEY_GUI_LEFT, HID_KEY_0,    0,                  HID_KEY_GUI_RIGHT,  0,                  0 },                //GPA3=103
  { HID_KEY_T,  HID_KEY_5,  HID_KEY_G,  HID_KEY_B,      HID_KEY_MINUS,  0,                  HID_KEY_BRACKET_LEFT, HID_KEY_F5,       0 },                //19
  { 0,          0,          0,          0,              HID_KEY_F10,    HID_KEY_ALT_RIGHT,  0,                  0,                  0 },                //GPB3=111
  { 0,          0,          0,          0,              HID_KEY_P,      HID_KEY_CAPS_LOCK,  0,                  0,                  0 },                //11
  { HID_KEY_U,  HID_KEY_7,  HID_KEY_J,  HID_KEY_M,      HID_KEY_SEMICOLON, HID_KEY_APOSTROPHE, 0,               HID_KEY_F7,         0 },                //GPB2=110
  { HID_KEY_I,  HID_KEY_8,  HID_KEY_K,  HID_KEY_COMMA,  0,              HID_KEY_ARROW_LEFT, HID_KEY_ARROW_RIGHT, HID_KEY_F8,        0 },                //18
  { HID_KEY_O,  HID_KEY_9,  HID_KEY_L,  HID_KEY_PERIOD, HID_KEY_BACKSLASH, HID_KEY_ARROW_DOWN, HID_KEY_ARROW_UP,  HID_KEY_F9,         0 },                //GPA5=105
  { 0,          0,          0,          0,              0,              0,                  0,                  0,                  HID_KEY_SHIFT_LEFT },   //14
  { 0,          0,          0,          0,              0,              0,                  0,                  0,                  HID_KEY_ALT_LEFT },     //GPA7=107
  { 0,          0,          0,          0,              0,              0,                  0,                  0,                  HID_KEY_CONTROL_LEFT }  //15
};
```

The above key matrix uses the key names required by the Adafruit TinyUSB library.

The Fn - media keys are not yet coded.